

Correcting a learned physical invariant improves world-model rollouts

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Abstract

Probing asks whether a model represents a quantity by fitting a readout to it. Its known weakness is that accuracy mixes what the representation stores with what the probe can learn, and the accepted remedy is better probe hygiene. We show the remedy addresses the wrong failure. In a DreamerV3 world model trained only on pendulum video, a polynomial readout fitted to *true* physical energy reaches a correlation of 0.9999 — yet the direction it identifies is one the model’s own transition does not preserve, and enforcing it during imagination makes the model’s physics worse. A label-free search that tracks energy slightly less well, but is $6.3\times$ better preserved by the transition, improves it by 41.8%. Whether a latent encodes a quantity and whether the dynamics preserve it are different properties, and they dissociate on four further axes — untrained models, out-of-distribution states, a second architecture, and an untrained transition — always in the same direction, with decodability surviving where conservation fails. A statistic on the transition operator separates them in every case. The quantity that statistic selects is causally deployed: enforcing it cuts drift in decoded physical energy by 55–76% on unseen trajectories, with 0 of 60 magnitude-matched random directions beating it; setting it steers the imagined world’s true energy (rank correlation 0.80–0.91); and the effect survives an interchange 50 steps into autonomous imagination. It is also bounded — to the training distribution, and to RSSM-like architectures.

1 Introduction

Probing is how we ask whether a model represents something. Fit a readout from the representation to the property of interest; if it succeeds, the property is there. The method’s known weakness is that probe accuracy mixes what the representation stores with what the probe can learn, and the accepted remedy is better probe hygiene — control tasks, selectivity, randomly initialised baselines [Hewitt and Liang, 2019, Belinkov, 2022].

We show that the remedy addresses the wrong failure. On a DreamerV3 world model trained only on pendulum video, we fit a polynomial readout to *true* physical energy. It reaches a correlation of 0.9999: as a probe it could not do better. Yet the direction it identifies is one the model’s own transition does not preserve, and enforcing it during imagination makes the model’s physics *worse*. A label-free search that tracks energy slightly less well, but is $6.3\times$ better preserved by the transition, improves it by 41.8%.

The failure is not in the probe. It is that whether a latent *encodes* a quantity and whether the model’s dynamics *preserve* it are different properties of the same representation, and no amount of probe methodology separates them — because the probe is already perfect.

Measuring the operator instead. The discriminating measurement is not on the representation but on the transition: apply one autonomous step at a real encoded state and ask how much a candidate scalar moves (Equation 4). This single statistic separates trained from untrained models ($74\times$), conservative from dissipative ($39\times$, no overlap), in-distribution from out ($55\text{--}267\times$), one architecture from another ($660\times$), and a label-free scalar from a supervised probe of the same quantity ($6.3\times$). On every one of those contrasts a probe reports success.

The dissociation is directional in each case we can construct: decodability is retained where conservation fails. Probing therefore over-reports, and it over-reports in the situations that matter — outside the training

distribution, in architectures that predict well without conserving, and when the quantity is supplied by supervision rather than found in the dynamics.

The quantity the operator selects is used by the model. A conserved-looking scalar could be an artifact of the search. Three interventions say otherwise. Projecting the latent back toward its level set during imagination reduces drift in *decoded physical energy* by 55–76% at long horizon on trajectories the method never saw, with 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beating it. Setting the scalar to an independent trajectory’s value moves the imagined world’s true energy toward that trajectory’s true energy (rank correlation 0.80–0.91, 0 of 75 controls). The same interchange applied 50 steps into an autonomous rollout — on a state the model reached by itself — works nearly as well, which is what a dormant pathway would not do [Makelov et al., 2024].

And it has boundaries worth stating. Outside the training energy band a freely fitted probe still recovers energy at 0.999 while the transition conserves it 55–267× worse: the representation generalises, the conservation does not. On a deterministic conv-GRU world model, identification transfers on 2 of 3 seeds and conservation on none, and training its transition seven times longer does not close the gap. The supportable scope is RSSM-like models, in distribution.

Contributions.

1. **A demonstration that probing over-reports, with a decisive control.** A supervised probe of the true physical quantity, at $|\rho| = 0.9999$, identifies a direction the dynamics do not use and that harms rollouts when enforced. The dissociation appears on four further independent axes and is directional in all of them.
2. **An operator statistic that discriminates where probes cannot,** together with the matched controls — magnitude-matched rather than norm-matched — that make its intervention interpretable. We show the norm-matched null used in prior work on this system controls nothing, because the correction is scale-invariant in the constraint.
3. **Causal evidence that the selected direction is deployed,** including an interchange 50 steps into autonomous imagination, and **a boundary condition:** the phenomenon is bounded to the training distribution and does not transfer to a deterministic recurrent architecture.

2 Recovering a latent invariant

Model and data. We analyze the published DreamerV3 architecture [Hafner et al., 2023]: 13.5M parameters, categorical 32×32 stochastic latents, a convolutional encoder and decoder, Kullback–Leibler balancing, and unimix. We train only the world model, with no actor or critic.

The data come from Gymnasium’s **Pendulum-v1** [Towers et al., 2024]. Each simulator step renders a 500×500 frame, which we crop to 448×448 and block-average to 64×64 . We set the simulator state directly, sampling $\theta \sim U[-1.8, 1.8]$ and $\dot{\theta} \sim U[-2.2, 2.2]$, and reject any trajectory that reaches the simulator’s $|\dot{\theta}| = 8$ speed clip, since clipping would break conservation. Actions are always zero. A single frame does not reveal velocity, so the model must infer $\dot{\theta}$ from the sequence.

We train on 204 trajectories of 120 frames using Adam at 10^{-4} , batch size 16, and sequence length 64, capped at 30 minutes of wall-clock time, which gives about 6,500 gradient steps. The remaining 52 trajectories are reserved as a post-training *analysis set*. Three independently trained seeds pass checks fixed before analysis: KL divergence above 1 nat, one-step decoding at least 4× better than predicting the dataset mean, and finite rollouts. Appendix A gives the remaining training details. Our claims concern a DreamerV3 world model trained on pendulum video, not DreamerV3 as a full reinforcement-learning agent.

Three kinds of latent trajectories. We use three latent trajectories and keep them separate throughout.

An *observation-conditioned* trajectory is the sequence of deterministic recurrent states produced when the model receives the next real frame at every step. This is what the encoder returns on the analysis set.

The *one-step transition* T is the model’s autonomous recurrent update with zero action and no new observation.

An *imagination rollout* starts from one encoded state and repeatedly applies T without supplying any later frames.

We estimate the latent flow by applying T once at each state of an observation-conditioned trajectory, so the transition is autonomous even though the states come from real video. Section 3 measures conservation along observation-conditioned trajectories. Section 4 tests what happens to the same scalar under autonomous imagination.

Latent coordinates. After training we freeze the model. Let h denote the deterministic recurrent state on the analysis set and $\bar{h}$ its mean. We take the top 12 principal directions of h , remove any direction with no support in the data, and let P denote the resulting orthonormal map. All extraction, projection, and perturbation operate on

$$z = P^\top(h - \bar{h}). \quad (1)$$

Let $F(z)$ be the projection through P of the model’s one-step displacement $T(h) - h$. Using the model’s own displacement avoids fitting a separate vector field and then analyzing that surrogate.

The candidate family. We look for the invariant among degree-4 polynomials in z . Writing $\varphi(z)$ for the vector of monomials up to degree 4 in the 12 latent coordinates, a candidate is $C(z) = a^\top \varphi(z)$, and the search is over the coefficient vector a . There are 1819 such monomials, from z_1 through $z_1 z_2 z_3 z_4$, excluding the constant term, which is conserved trivially. The pendulum’s energy is quadratic in $\dot{\theta}$ and needs polynomial terms to approximate $\cos \theta$, and the latent is an unknown nonlinear encoding of $(\theta, \dot{\theta})$ rather than those coordinates themselves, so the family must be rich enough to express energy through that distortion while staying small enough to fit.

The invariance criterion. An invariant is constant along any one trajectory and differs between trajectories. We score a candidate by

$$\text{ratio}(C) = \frac{\text{mean within-trajectory variance of } C}{\text{total variance of } C}, \quad (2)$$

which is 0 for a perfectly conserved scalar and near 1 for one that wanders as much within a trajectory as across the dataset. The denominator keeps the criterion non-trivial: without it, $C = 0$ would score perfectly while distinguishing nothing.

Both variances are quadratic forms in a . With W the mean within-trajectory covariance of the features and T their total covariance, Equation 2 is $a^\top W a / a^\top T a$, a Rayleigh quotient. Its minimiser is the eigenvector of the generalized problem $W a = \lambda T a$ with the smallest eigenvalue, and λ is the invariance ratio itself. One eigendecomposition returns the whole family, ranked from most to least conserved.

Selecting among conserved candidates. Conservation alone does not pick out a unique scalar. If C is conserved then so is any function of it, so E , E^2 and mixtures of E with other conserved quantities all sit near the top of the ranking, and the leading eigenvector is generally some combination of them. We therefore use flow alignment as a secondary criterion among the leading candidates, fitting C jointly with an antisymmetric operator B such that

$$F \approx B \nabla C, \quad B^\top = -B. \quad (3)$$

This mirrors the Hamiltonian relation between a conserved quantity and the flow it generates: an anti-symmetric operator maps the gradient of C to a direction tangent to its level set [Arnold, 1989]. Because $\nabla C^\top B \nabla C = 0$ for any antisymmetric B , a scalar satisfying Equation 3 is constant along the flow by construction. The criterion also separates candidates the invariance ratio cannot: E^2 is exactly as conserved as E but does not pair with the same B .

Table 1: The correct physical quantity, fitted optimally, is worse conserved and harms rollouts. Three independently trained models; 20 magnitude-matched random constraints per model.

constraint	$ \rho _E$	ρ_{obs}	change in energy drift
label-free C	0.930–0.973	6.3–8.7 $\times 10^{-3}$	– 41.8%
supervised energy probe	0.9999	4.6×10^{-2}	+20.0%
random, magnitude-matched	~ 0.12	~ 1.0	+6.4%

The fit is bilinear, so fixing either a or B makes the other a least-squares problem and we alternate. It searches within the top eight eigenvectors from Equation 2, so the recovered C has eight effective degrees of freedom rather than 1819. That matters for a search run on 52 trajectories, where a free fit over the full basis could drive the in-sample ratio to zero by overfitting. Appendix A isolates the contribution of the flow criterion: conservation drives most of the recovery and intervention effect, while flow alignment mainly reduces variation across seeds.

Reference energy. We compare C with the textbook pendulum energy $E = \frac{1}{6}\dot{\theta}^2 + 5 \cos \theta$. Gymnasium uses a semi-implicit integrator, so it conserves a nearby shadow Hamiltonian $\tilde{H} = H + O(\Delta t)$ rather than E itself [Hairer et al., 2006a]. In our trajectories this produces about 12% relative oscillation in E without secular drift, which we treat as a noise floor.

The search sees only latent states and the model’s own one-step transition. It never sees θ , $\dot{\theta}$ or E , and nothing in Equations 2 and 3 refers to them. Ground truth enters afterward, when we score the recovered C against E . We selected the extraction dimension on three development models and fixed it before evaluating the three models reported here.

Code and data. The extraction code, the run logs behind every figure and table, and the source of this paper are at <https://github.com/Zarand3r/world-model-invariants>. The figures regenerate from the committed logs without a GPU.

Preliminary experiments. We developed the extraction procedure in preliminary experiments on smaller recurrent models. Those experiments motivated the untrained, dissipative, and intervention controls used here, and we do not use them as evidence for the DreamerV3 results.

3 Decodability is not dynamical structure

A probe establishes that a quantity can be read out of a representation. It says nothing about whether the model’s own transition uses or preserves that quantity. We measure the second property directly, with a statistic on the transition operator: apply one autonomous step at a real encoded state and ask how much a candidate scalar changes,

$$\rho_{\text{obs}}(C) = \frac{\text{median}_t |C(T(z_t)) - C(z_t)|}{\text{std}_{\text{traj}} C}, \quad (4)$$

normalised so it is invariant to the scale of C and comparable across candidates. A value near zero means the transition preserves C ; a value near one means a single step moves C by its entire across-trajectory spread.

A perfect probe identifies a direction the dynamics do not use. The sharpest test replaces the label-free search with the answer it is trying to find. We fit a degree-4 polynomial readout to *true* pendulum energy by ridge regression on the same latent, then use it to drive the identical correction. It reaches $|\rho|_E = 0.9999$: as a probe it is essentially perfect.

Table 1 shows the outcome. The supervised probe tracks energy better than the label-free scalar on every seed, is $6.3\times$ less preserved by the transition, and *increases* rollout energy drift where the label-free scalar reduces it by 41.8%. All three comparisons hold on 3 of 3 models. The direction that best predicts the physical variable and the direction the model’s dynamics actually respect are not the same direction.

Table 2: Decodability is retained while conservation degrades, across four independent axes. ρ_{obs} for the trained conservative model in-distribution is $6.3\text{--}8.7 \times 10^{-3}$.

axis	contrast	decodability	ρ_{obs} degradation
training	untrained network	$ \rho _E$ up to 0.908	$74\times$
state space	outside training energies	free probe 0.999	$55\text{--}267\times$
architecture	deterministic conv-GRU	$ \rho _E$ 0.89–0.97	$660\times$
objective	transition never trained	$ \rho _E$ 0.97	$\sim 10\,000\times$

This also answers the practical objection to a label-free method. The search is not a workaround for missing labels: it finds something a supervised fit on perfect labels does not, because it optimises conservation by the transition rather than agreement with the variable.

The dissociation is directional, and appears on four further axes. In every case we can construct, decodability survives where conservation fails; we have not found the converse. Table 2 collects them. On each row a probe would report success.

The out-of-distribution row is the cleanest, because it holds the model fixed and varies only the region of state space. Outside the training energy band a freely fitted probe recovers energy at 0.999 — the encoder generalises — while the transition conserves it $55\text{--}267\times$ worse, reaching values statistically indistinguishable from a random constraint at the earliest checkpoint. Decodability and conservation come apart inside a single trained model, split by where in state space it is asked.

Conservation, not decodability, predicts the intervention. Among candidate scalars matched in energy correlation and spanning five orders of magnitude of conservation quality, repair magnitude tracks conservation (Spearman $+0.71$ and $+0.60$ on two models). We report a limitation of this control rather than resolve it: conservation and decodability remain rank-correlated even inside a band constructed to hold decodability fixed (-0.62 and -0.64), and that residual correlation reproduces across independently trained models. On a system whose transition conserves energy, the well-conserved latent directions *are* the energy-like ones, which is why a matched band can only be built at low decodability and why the two properties may not be fully separable by this design.

4 The privileged direction is causally deployed

If the recovered scalar is only a descriptive correlate, enforcing it has no reason to help and setting it has no reason to steer. Three interventions test the alternative, each with matched controls.

A magnitude-matched null, and why the obvious one fails. The natural correction projects the latent back toward the level set, $z \leftarrow z - \alpha(C(z) - C_0) \nabla C / \|\nabla C\|^2$. This is a Newton step, so its size grows with how badly the constraint is violated — and a random constraint is violated far more than a conserved one. Matching random draws in *coefficient* norm, as one might expect to control for scale, controls nothing: the step is invariant under $C \rightarrow \lambda C$. Measured, random draws take steps $29\times$ larger than the recovered constraint, so the comparison conflates direction with magnitude.

We therefore fix the step size and vary only direction, $z \leftarrow z - \varepsilon \text{sign}(C - C_0) \nabla C / \|\nabla C\|$, over a frozen ε grid. Arms then differ in where they push, never in how hard.

Correcting the invariant repairs the imagined physics. We evaluate on decoded physical energy, not pixels: a non-learned geometric readout recovers θ from the rendered frame by image moments and θ by backward differences, matching the simulator’s semi-implicit convention, with a median decoded-energy error of 1.6% of the across-trajectory spread. The statistic is the secular drift of decoded energy over a rollout.

At $H = 100$ the recovered direction reduces drift by 32–51% across three models. On 512 trajectories from a disjoint generator seed, with C and the entire coordinate frame frozen, the effect does not shrink:

−54.6%, −58.8%, −75.9% at $H = 190$. Across both horizons **0 of 60** magnitude-matched random directions and **0 of 15** equal-norm tangent directions beat the recovered one on any model.

Setting the invariant steers the imagined physics. Restoring C shows drift is costly; it does not show C is a control variable. We take a recipient state and an independent donor trajectory, move the recipient minimally along the local C -normal until $C(z) = C(z_{\text{donor}})$, and let the rollout run free. The intervention never sees true energy — the target is read from the model’s own latent.

The imagined world’s *true decoded* energy moves toward the donor trajectory’s *true* energy, with rank correlation 0.802–0.914 across three models and **0 of 75** random and tangent controls beating it. A preregistered offset sweep is monotone in both directions on every model.

The subspace is used by the model’s own dynamics, not only reachable from encoder states.

A subspace edit can produce the expected output change through a pathway the model does not normally use [Makelov et al., 2024]. We therefore repeat the donor interchange *50 steps into an autonomous rollout*, on a state the model reached by itself rather than one the encoder produced.

The effect barely degrades: rank correlation 0.762–0.849 at depth 50 against 0.803–0.924 at depth 0, with 0 of 13 controls beating it at either depth on any model. Equal-norm tangent edits, which cannot change C to first order, sit near zero at both depths. A dormant pathway would lose its effect precisely where this test applies it.

The same pipeline refuses when there is nothing to find. On matched dissipative models, where no nontrivial first integral exists, the extraction returns a direction that is not distinguished from a random one: 24 of 60 random draws beat it, and it harms rollouts on 2 of 3 models. Its operator statistic sits at $3.4\text{--}3.7 \times 10^{-1}$, in the same range as an untrained network and $39\times$ worse than any conservative model, with no overlap between the two groups.

5 Where the phenomenon stops

Both boundaries below are measurements with matched controls rather than caveats, and each constrains what produces conservation.

The invariant is bounded to the training distribution. We generate trajectories whose energies lie outside the training band, holding the integrator, renderer and rejection rule fixed. The frozen C collapses there, $|\rho|_E$ falling from 0.93–0.97 to 0.03–0.31.

Refitting the entire pipeline on those latents separates two explanations. Identification recovers immediately — $|\rho|_E$ of 0.92–0.96, and a freely fitted probe reaches 0.999 — so the information is present and the polynomial family can express it. What does not recover is conservation: ρ_{obs} is 55–267 $\times$ worse than in-distribution. There is no well-conserved quantity to find outside the training band, which is why a scalar fitted where conservation held has nothing to lock onto.

The model’s *representation* generalises out of distribution. Its *conservation* does not.

The phenomenon is architecture-dependent. We repeat the full pipeline on a deliberately different world model: a deterministic convolutional GRU autoencoder with no stochastic latent, no KL term and no unimix, its recurrent state matched at 512 so the extraction operates on a latent of the same dimensionality, and its transition trained by an open-loop rollout term. The analysis code runs unchanged.

Identification transfers on 2 of 3 models ($|\rho|_E$ 0.97, 0.89, 0.19). Conservation transfers on none: ρ_{obs} is 1.75–9.02 against the RSSM’s $\sim 7 \times 10^{-3}$. Training the transition seven times longer did not close the gap.

We are explicit about what this comparison cannot settle. An open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths, so the residual gap narrows toward the architecture without isolating it. What the comparison does establish is that conservation is not an automatic consequence of training a pixel world model whose transition rolls out accurately, and the supportable scope of every claim above is RSSM-like models, in distribution.

6 Limitations

Two physical systems, both smooth and Hamiltonian: a pendulum and a two-degree-of-freedom anharmonic oscillator. No contacts, no multiple interacting objects, no actions. Whether any of this survives discontinuous dynamics is untested.

Three trained models per arm. We report ranges rather than confidence intervals for model-level statistics throughout, because three is too few for a precise effect size, and we take the model seed rather than the trajectory as the independent unit for every model-level claim.

The extraction searches a degree-4 polynomial family in a 12-dimensional principal subspace. The invariance ratio it minimises *is* the normal component of one-step error, so the recovered scalar having low normal error is the objective being met rather than a finding; the empirical content is that the resulting direction coincides with physical energy, which Section 3 shows supervision does not reproduce.

The architecture comparison does not isolate architecture. An open-loop reconstruction term and a prior-posterior KL are different losses, not the same loss at different strengths, so the conv-GRU result narrows the explanation toward the latent design without establishing it.

We show no downstream consequence. Whether invariant drift predicts rollout failure usefully, or whether the correction helps a planner, is untested, and the effect on pixel error alone at short horizon is small (1.7% at $H = 50$). The large effects reported here are on decoded physical energy, a metric we introduce because it measures the property in question.

Finally, the two-invariant case exposes a limitation of the extraction rather than of the model. On a system conserving both energy and angular momentum, the conserved subspace reliably contains both (angular-momentum correlation 0.94–0.98 in every measurement) while the joint fit isolates either in only 1 of 7 measurements across three models. The flow-alignment criterion, which selects a single direction inside the conserved subspace, has no unique answer when two independent invariants exist.

7 Related work

What probes establish. Probing shows a variable is recoverable from a representation, not that the model’s computation uses it [Alain and Bengio, 2017]. Control tasks [Hewitt and Liang, 2019] and later critiques [Belinkov, 2022] address the confound between what the representation stores and what the probe can learn. Our result is a different failure: the probe in Section 3 is fitted to the true physical quantity and reaches 0.9999, so no control-task correction applies to it, and it still identifies a direction the dynamics do not use. Work on emergent world representations goes beyond probing by intervening [Li et al., 2023, Nanda et al., 2023], which is the logic our interventions follow; Makelev et al. [2024] show such interventions can succeed through pathways the model does not normally use, which is the objection our depth-50 interchange is designed to answer.

Discovering conserved quantities. A substantial literature finds conservation laws from data. Con-servNet minimises a noise-regularised within-group variance [Ha and Jeong, 2021], of which our invariance ratio is the closed-form Rayleigh-quotient analogue; AI Poincaré counts conserved quantities from trajectory manifolds [Liu and Tegmark, 2021]. ConCerNet learns a conservation law and then projects the learned dynamics onto it [Zhang et al., 2023], and FINDE finds first integrals and preserves them by projection [Matsubara and Yaguchi, 2023]; Noether Networks meta-learn a conserved quantity and apply it at test time, including on pixel pendulums [Alet et al., 2021]. All of these *impose or co-train* conservation. We ask a different question: whether a model trained with no such term has a scalar its transition already preserves, and we measure that on the frozen transition rather than building it in.

Structure by design, and whether it is what helps. Hamiltonian and Lagrangian networks build conservation into an architecture [Greydanus et al., 2019, Cranmer et al., 2020]. Gruver et al. [2022] find that HNN generalisation comes from modelling acceleration and coordinate choice *rather than* from symplectic structure or energy conservation — which licenses the objection that our projection is a generic regulariser. Our magnitude-matched null answers it directly: with step size fixed, 0 of 60 random directions and 0 of 15

equal-norm tangent directions match the recovered one, and on dissipative models the same pipeline returns a direction indistinguishable from random.

Drift in learned dynamics. That accurate one-step models drift over long rollouts is documented across fields. In machine-learned force fields, force accuracy and simulation stability decouple sharply [Fu et al., 2023]. In numerical analysis, projection onto a level set is a standard structure-preserving device [Hairer et al., 2006b], and our correction is that device applied to a *discovered* rather than a given invariant, in a learned latent rather than the physical state.

Evaluating physics in world models. World models are normally evaluated by prediction quality or control performance [Hafner et al., 2023, Ha and Schmidhuber, 2018, Schrittwieser et al., 2020], and physics benchmarks evaluate generated output. Neither reads the model’s internal state. The disentanglement literature [Locatello et al., 2019] documents how readily an appealing structural claim about representations survives without the control that would falsify it.

8 Conclusion

A probe fitted to the correct physical quantity, at a correlation of 0.9999, identifies a direction a world model’s dynamics do not use — and enforcing it degrades the model’s imagined physics, while a label-free scalar that is better conserved and slightly less accurate improves it. Decodability and dynamical structure are different properties of a representation, they come apart in every setting we could construct, and only a measurement on the transition operator tells them apart.

The practical consequence is narrow and concrete. Finding that a physical variable is decodable from a world model’s latent says little about whether the model’s rollouts respect it, and it says least in the situations that matter most: outside the training distribution, in architectures that predict well without conserving, and when the variable is supplied by supervision rather than found in the dynamics. Reporting a probe result in those settings will overstate what the model has learned. The operator statistic costs one autonomous step per state to compute.

A Experimental details and extraction ablations

Training. We set free bits to 0 rather than DreamerV3’s reference default of 1.0. The measured KL divergence settles near 1.8 nats, well above collapse, so the floor is inactive here. All three trained models pass the acceptance checks; none were discarded.

Data split. Training uses trajectories 0–203 of the 256-trajectory dataset. The remaining 52 form the analysis set. We fit C on those 52 trajectories and evaluate the Section 4 correction on the same 52, so the absolute effect is in-sample with respect to invariant fitting. The recovered and random-constraint arms use the same trajectories.

Scoring the correction over a grid. We score the slope of rollout error over the fixed α grid rather than the minimum on that grid. Taking the minimum selects the most favourable of five points after seeing the curve. In one diagnostic run that rule would have reported a 5.9% improvement for an arm whose error rose monotonically to twice baseline.

Flow alignment contributes reproducibility. Holding the candidate family, data, dimension, degree, and α grid fixed, we select C either by the invariance criterion alone or by the joint criterion. Conservation alone recovers energy at 0.950 against 0.973 for the joint rule, and gives a slightly better mean intervention slope (-0.087 against -0.077 , winning on two of three models). The difference is in spread: recovery varies by 0.035 across seeds under conservation alone and by 0.008 under the joint rule.

The pairing residual does not track recovery. As the extraction dimension rises from 6 to 12, energy recovery improves from 0.721 to 0.967 while the pairing residual worsens from 0.738 to 0.876. Recovery already reaches 0.967 at dimension 8, where the residual is lower at 0.813. The added dimensions do not carry energy while resisting the Hamiltonian fit: directions 7–12 account for 44.9% of the residual and 43.5% of the flow. Figure 2 gives the sweep.

B Latent sensitivity analyses

For each direction u of the extracted subspace we measure its variance and the damage a displacement along it does to an imagination rollout,

$$V(u) = \text{Var}(u^\top z), \quad D_H(u) = \mathbb{E}[L_H(z + \epsilon u) - L_H(z)], \quad (5)$$

with L_H the H -step imagination error against the analysis set.

Calibrating the displacement. At $\epsilon = 0.05|z|$ the damage is about -0.5% of baseline with the sign varying across directions. Imagination is deterministic, so repeated unperturbed rollouts are identical and these values are not sampling noise. Damage for the leading direction runs -0.5% , $+3.1\%$, $+18.1\%$, $+46.4\%$, and $+128.8\%$ at $\epsilon/|z|$ of 0.05, 0.10, 0.25, 0.50, and 1.00. We use $\epsilon = 0.25|z|$, the smallest displacement whose damage exceeds 10% of baseline while the response still grows smoothly.

A displacement does not decay under imagination. After 40 steps it retains $6.3\times$ its initial size at $\epsilon = 0.05|z|$, $2.8\times$ at $0.25|z|$, and $1.9\times$ at $0.5|z|$, as medians over the three models.

Variance and sensitivity at different horizons. At a 40-step horizon the two rankings anticorrelate on all three models (-0.364 , -0.259 , -0.406), with damage spanning $13.5\times$ to $118.8\times$ between the most and least consequential direction (Figure 1a). The ranking replicates: split-half rank correlation of D_H has median 0.95 across the settings we tried, with a minimum of 0.52 at the smallest displacement and longest horizon. The disagreement with variance does not extend to long horizons. Pooling three models and three displacements, median $\rho(V, D_H)$ is -0.252 at horizon 20, negative in 9 of 9 settings; -0.196 at horizon 50, negative in 5 of 9; and -0.028 at horizon 100, negative in 5 of 9, with one setting reaching $+0.727$. Once a rollout has diverged, high-amplitude directions dominate the error and the rankings realign.

The correction pushes along higher-damage directions at $+0.385$, $+0.692$, and $+0.287$ (Figure 1b), without having been given that ranking. The correlations stay well below 1, so the correction is only partly concentrated there. All of this is measured on the conservative models.

Basis-independent properties reproduce better than coordinate selectors. The three models agree closely on quantities that need no choice of basis: energy recovery varies by 0.008, intervention slopes by 0.016, and the top-3 eigenvalue mass of $M_C = \mathbb{E}[gg^\top / \|g\|^2]$ with $g = \nabla C$ by 0.046 (0.448, 0.441, 0.487). Selectors on individual coordinates are less stable: ranking coordinates by ∇C and keeping the top three gives intervention slopes of -0.075 , $+0.016$, and -0.089 , beating a variance ranking on two models and changing sign across seeds. An orthogonal projector onto the top eigenspace of M_C beats the variance selector on all three models, though on one it sits at the 41st percentile of 100 random rank-3 subspaces. Since the correction is rank one at each state, the spectrum of M_C measures how much ∇C rotates as the state changes, and the amount is similar across models. This is consistent with the models encoding the same scalar in different latent coordinate systems, although we do not directly align their representations.

C Additional sweeps

Energy outside the highest-variance directions. On two of three models, principal directions 7–12 carry about twice the energy information of the top six ($2.23\times$ and $2.13\times$); on the third the ratio reverses to 0.91. We report the models separately because a median would hide the sign change. Figure 3 shows the decomposition.

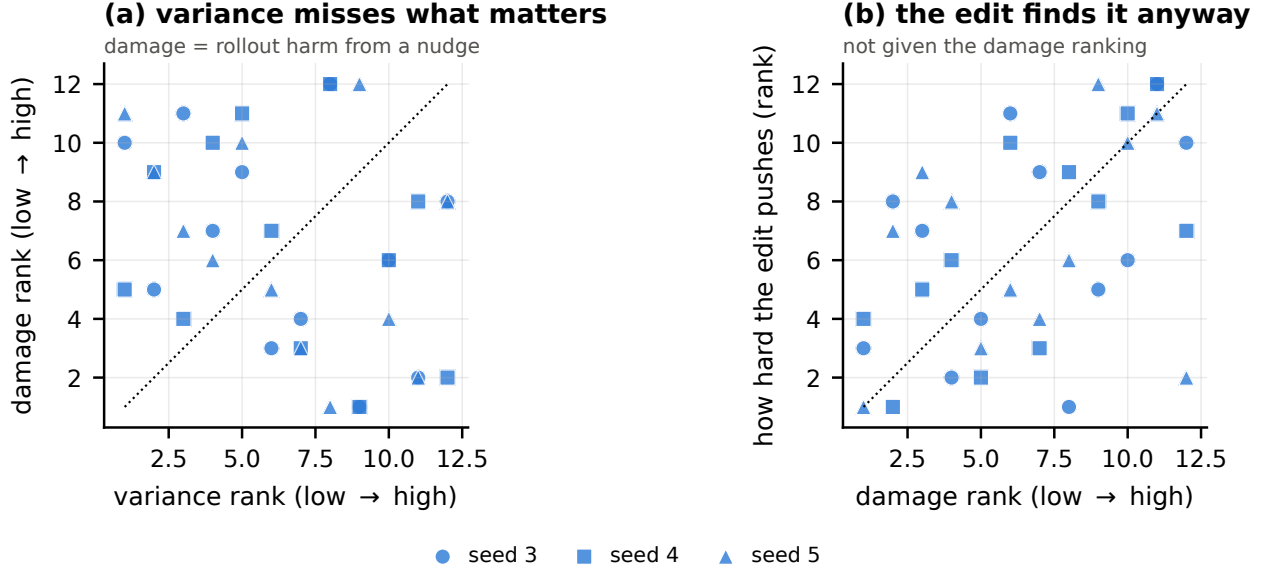


Figure 1: Each point is one of the 12 extracted directions, ranked within its own model; the dotted line marks equal ranks. **(a)** Variance rank against damage rank at a 40-step horizon. **(b)** Damage rank against the magnitude of the invariant correction along that direction.

Accumulated invariant error. Accumulated invariant error predicts which rollouts degrade most: an integral predictor reaches $R^2 = 0.187$ against 0.021 for a fitted power law in time at the horizon where both perform best, and shuffling destroys the relation. The current experiments do not distinguish weighted from unweighted accumulation. In a nested comparison with 400 bootstrap resamples, the weighted term adds explanatory power beyond the unweighted term in 1 of 9 seed-horizon combinations, and the unweighted beyond the weighted in 0 of 9, at $n = 512$.

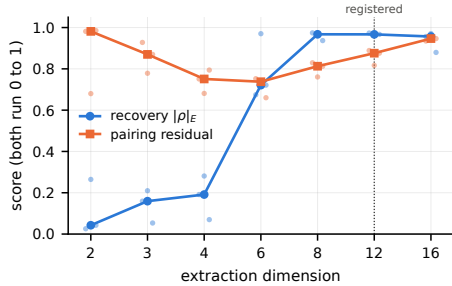


Figure 2: Energy recovery and pairing residual against extraction dimension. Lines are medians over the three models; points are individual models. Both quantities are dimensionless.

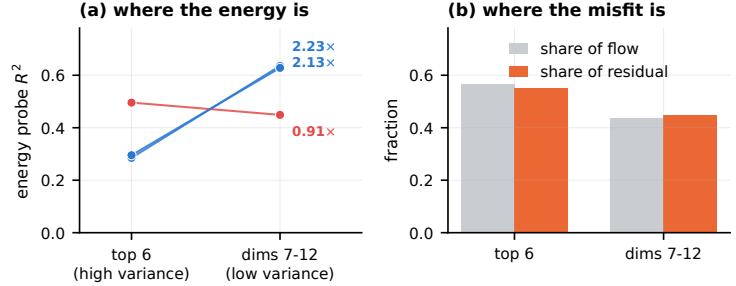


Figure 3: **(a)** Energy-probe R^2 for the top six principal directions against directions 7–12, per model. **(b)** Fraction of latent flow and pairing residual carried by each group, medians over three models.

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